

The schematic diagram illustrates the SAR868 VHF transmitter circuit, which includes two main sections: SAR868 UHF and SAR868 VHF.

SAR868 UHF Section:

- Powered by +3.7V and GND.
- Includes components C21 (10µF), C22 (10nF), R9 (10K), and R10 (H/L).
- Microcontroller U17 (SAB8U) has pins ANT, MIC_IN, AF_OUT_UHF, SQ, PTT, PS, RXD, TXD, and GND connected.
- Outputs include AF_OUT_UHF, SQ, PTT_UHF, PS, RXD_UHF, and TXD_UHF.

SAR868 VHF Section:

- Powered by +3.7V and GND.
- Includes components J27 (Conn_Casual), R1k, LED, C20 (10µF), and C26 (10nF).
- Microcontroller U10 (SAB8V) has pins ANT, MIC_IN, AF_OUT_VHF, SQ, PTT, PS, RXD, TXD, and GND connected.
- Outputs include AF_OUT_VHF, SQ, PTT_VHF, PS, RXD_VHF, and TXD_VHF.

Power Regulation:

- A +5V input is regulated down to +3.7V using a diode D1 and capacitors C29 (10µF) and C30 (10nF).

74HCT125 MODULE:

- Powered by +5V and GND.
- Inputs include VCC, GND, 1A, 1B, 1C, 1D, 2A, 2B, 2C, 2D, 2Y, and 2Z.

NEO-6M GPS:

- Powered by +3.3V and GND.
- Inputs include VCC, GND, RX_GPS, TX_GPS, TX, TX_ZT, and TX_ZT.

Connectors:

- J26 (Conn_Casual) connects to the SAR868 UHF section.
- J27 (Conn_Casual) connects to the SAR868 VHF section.
- J1 (Conn_Sisx8) and J3 (Conn_Sisx8) are multi-pin connectors for external modules.

The diagram illustrates the hardware setup for a heart rate monitor. It features an **ARDUINO_NANO** connected to an **MT8870 DTMF** module and an **S15351** module. The **MT8870 DTMF** module is connected to the **ARDUINO_NANO** via I2C (SDA, SCL) and UART (TX, RX). The **MT8870 DTMF** module is also connected to a 5V supply and ground. The **S15351** module is connected to the **MT8870 DTMF** module via I2C (SDA, SCL) and to an SDA/SCL header. The **S15351** module is also connected to a 3.3V supply and ground. The diagram includes labels for various components and their connections, such as TX, RX, SDA, SCL, and GND.

The diagram illustrates the PCB layout for the ESP32-DEVKIT-V1, showing the placement of various components and their connections. The board is populated with an ESP32-DEVKIT-V1 module, a W290 module, a MCP23017 I2C module, a LORA module, a CD4066 Switch, and a FILTER module. The layout includes power planes, ground planes, and various passive components like capacitors and resistors. The components are labeled with their respective pin numbers and functions. The board is shown in a top-down view with a white background and black text.

The schematic diagram illustrates the electrical connections for the INA219 module. It includes the following components and connections:

- INA219 Module (U4):**
 - VCC:** Connected to +3.3V through a 10µF capacitor (C10) and to GND through a 10µF capacitor (C9).
 - VS:** Connected to +3.3V.
 - SHED:** Connected to GND.
 - VM:** Connected to the VM pin of the DS90C01 module (U3).
 - SL:** Connected to the SCL pin of the DS90C01 module (U3).
 - SDA:** Connected to the SDA pin of the DS90C01 module (U3).
 - SHIN:** Connected to the SHIN pin of the DS90C01 module (U3).
 - SHOUT:** Connected to the SHOUT pin of the DS90C01 module (U3).
 - SHIN+:** Connected to the SHIN+ pin of the DS90C01 module (U3).
 - SHOUT+:** Connected to the SHOUT+ pin of the DS90C01 module (U3).
- DS90C01 Module (U3):**
 - VCC:** Connected to +3.3V.
 - VS:** Connected to +3.3V.
 - SHIN:** Connected to the SHIN pin of the DS90C01 module (U3).
 - SHOUT:** Connected to the SHOUT pin of the DS90C01 module (U3).
 - SHIN+:** Connected to the SHIN+ pin of the DS90C01 module (U3).
 - SHOUT+:** Connected to the SHOUT+ pin of the DS90C01 module (U3).
- 12-pin Header:**
 - Pin 1:** Connected to GND.
 - Pin 2:** Connected to +3.3V.
 - Pin 3:** Connected to +3.3V.
 - Pin 4:** Connected to +3.3V.
 - Pin 5:** Connected to +3.3V.
 - Pin 6:** Connected to +3.3V.
 - Pin 7:** Connected to +3.3V.
 - Pin 8:** Connected to +3.3V.
 - Pin 9:** Connected to +3.3V.
 - Pin 10:** Connected to +3.3V.
 - Pin 11:** Connected to +3.3V.
 - Pin 12:** Connected to +3.3V.

The schematic diagram illustrates the power and signal connections for the ESP32-CAM module. The module is represented by a yellow rectangle with pins on all sides. The connections are as follows:

- Power Connections:**
 - +5V:** Connected to the module's +5V pin and the +5V pin of the breadboard.
 - GND:** Connected to the module's GND pin and the GND pin of the breadboard.
 - +3.7V:** Connected to the module's +3.7V pin and the +3.7V pin of the breadboard.
 - 5V:** Connected to the module's -5V pin and the -5V pin of the breadboard.
- Signal Connections:**
 - PWR_FLAG:** Connected to the module's PWR_FLAG pin and the PWR_FLAG pin of the breadboard.
 - SSVT_DONE:** Connected to the module's SSVT_DONE pin and the SSVT_DONE pin of the breadboard.
 - TRIGGER_CAM:** Connected to the module's TRIGGER_CAM pin and the TRIGGER_CAM pin of the breadboard.
 - FILTER_IN:** Connected to the module's FILTER_IN pin and the FILTER_IN pin of the breadboard.
- Other Connections:**
 - Battery:** A battery (BT1, BT2) is connected to the module's +5V and GND pins.
 - USB Cable:** A USB cable (J1, J2) is connected to the module's USB pins.
 - Breadboard:** The breadboard contains various components, including resistors, capacitors, and a USB cable.